

SECTION B: 40 marks.

Each question weighs 10 marks

A 40-year-old male was found on the floor unresponsive by his neighbors was rushed to the hospital. He recovered within 30 minutes but was delirious and with deep labored breathing. You inquire from his best friend he tells you, the patient called him, two days ago to ask to be exempted from work because he had fever and he went to a health facility and was given medications and went back home but he does not know what he was treated for. As you are talking to his friend, the nurse reports to you the vital signs as follows: BP 107/69mmHg, PR=105bpm, RR= 22 cycles/min, SPO2=96% on RA. On exam his tongue is bitten on the sides and he is jaundiced. She later also debriefs she inserted NGT and Foley catheter which is now draining coca cola urine of around 70mls for the past 4 hours despite 500mls of NS which she secretly gave without your approval.

- a. What is the most likely diagnosis? And why (2marks)
- b. What are the possible differential diagnoses? (1marks)
- c. What are the important 4 physical signs you would like to know in this patient? (2marks)
- d. How will you manage him? (5marks)

2. A patient presents with purple macules and nodules over the legs. As you examine him you note that there is woody edema. He reports blood in stools occasionally.

- a. What is your diagnosis?
- b. What is the etiological agent associated with the disease?
- c. List five typical forms/types of the disease condition and how each presents briefly.
- d. List principles of management for the disease.

3. A patient presented with

2. A 30-year-old man complains of a rash on his face, particularly on the central part of his forehead, nose, and cheeks. You observe erythematous papules and pustules with some comedones. What is the most likely diagnosis?
- A. Acne vulgaris
 - B. Rosacea
 - C. Seborrheic dermatitis
 - D. Systemic lupus erythematosus (SLE)
 - E. Perioral dermatitis
3. A 25-year-old hiker comes to your clinic with a red, circular rash with a clear center. She reports that it started as a small red bump and expanded over time. The rash is not pruritic. What is the most likely diagnosis?
- A. Tinea corporis
 - B. Erythema multiforme
 - C. Lyme disease (Erythema migrans)
 - D. Pityriasis rosea
 - E. Urticaria
4. A 55-year-old man presents with silvery-white scaly plaques on his knees, elbows, and scalp. The affected areas are well-demarcated and non-pruritic. What is the most likely diagnosis?
- A. Psoriasis
 - B. Eczema
 - C. Lichen planus
 - D. Seborrheic dermatitis
 - E. Contact dermatitis
5. A 20-year-old college student visits your clinic with complaints of intense itching at night, particularly in the fingers, wrists, and groin area. On examination, you find linear burrows and vesicles. What is the most likely diagnosis?
- A. Scabies
 - B. Dermatitis herpetiformis
 - C. Pemphigus vulgaris
 - D. Bullous pemphigoid



- a. With reasons, what is the most likely diagnosis?
- b. List down three possible risk factors for the development of the diagnosis above
- c. List five possible groups of organisms that can cause such as presentation with a possible example for each group
- d. List treatment options for the diagnosis above

4. Briefly discuss

- i. The different types of scabies (3 marks)
- ii. How the diagnosis of scabies can be reached (2 marks)
- iii. The treatment options for scabies (5 marks)

Which one of the following statements regarding scabies is false?

- A. All members of the household should be treated
- B. Typically affects the fingers, interdigital webs and flexor aspects of the wrist in adults
- C. Scabies causes a delayed type IV hypersensitivity reaction
- D. Patients who complain of pruritus 4 weeks following treatment should be retreated
- E. Malathion is suitable for the eradication of scabies

7. A 54-year-old woman with a history of type 1 diabetes mellitus presents with unsightly toenails affecting the lateral three nails of the left foot. On examination the nails are brown and break easily. Nail scrapings demonstrate *Trichophyton rubrum* infection. What is the treatment of choice?

- A. Oral terbinafine for 12 weeks
- B. Oral itraconazole for 4 weeks
- C. Topical itraconazole for 2 weeks
- D. Topical amorolfine for 6 weeks
- E. Oral itraconazole for 1 week

8. A 23-year-old man presents as he is concerned over recent hair loss. Examination reveals a discrete area of hair loss on the left temporal region with no obvious abnormality of the underlying scalp. What is the most likely diagnosis?

- A. Telogen effluvium
- B. Alopecia areata
- C. Tinea capitis
- D. Male-pattern baldness
- E. Discoid lupus erythematosus

9. A 45-year-old male presents to ER with weight loss, sudden loss of appetite, abdominal pain, excessive fatigue, and joint pains. He also complains about lumps on the skin as well. His blood pressure was 170/100mmHg, pulse was 87/minute, temperature 38°C, and

respiratory rate 20/minute. Laboratory tests revealed creatinine 2.2 mg/dL, raised ESR, c-ANCA, AND p-ANCA are negative. Arteriography shows microaneurysms of the renal arteries. Which of the following is the treatment in this patient?

- A. Cyclophosphamide+ methotrexate
- B. Steroids +cyclophosphamide
- C. Aspirin
- D. Intravenous immunoglobulin
- E. Rituximab

10. A 32-year-old young male presented to ER with red teary eyes rash on his legs., he complains of abdominal pain, nausea, and diarrhea. He has a history of DVT twice, superficial phlebitis, and recurrent mouth ulceration. On examination of the eye, uveitis was seen. Which of the following is a gold standard treatment for the condition?

- A. Methotrexate
- B. Colchicine and NSAIDS
- C. Systemic steroids and azathioprine
- D. Cyclophosphamide
- E. Colchicine alone

11. A 70-year-old female presents to OPD with bluish discoloration of fingers, hematuria, heartburn, fatigue, and joint pains off and on. Which of the following tests will help establish the diagnosis?

- A. Chest X-ray
- B. MRI
- C. Doppler of the fingers
- D. CT scan of the chest
- E. Lung biopsy

12. A 40-year-old female comes to OPD with dry eyes and dyspareunia for the last 6 months. She also complains of cough and fatigue with joint pains. On examination, her parotid gland was enlarged and laboratory tests revealed anti-Ro antibodies are positive. Which of the following tests is recommended to this patient?

- A. Head CT

C. Schirmer test

D. Biopsy of the labia

E. Colposcopy

13. A 60-year-old female presents in OPD with knee joint stiffness in the morning and increases with activity and decreases on rest. She also complains about a crackling noise on joint movement. x-ray shows narrowing of the joint space and osteophytes. Which of the following treatments is recommended in this patient?

A. Methotrexate

B. Steroids

C. Acetaminophen

D. Elastic socks

E. Surgery

14. A 26-year-old female presented to ER with dry cough and shortness of breath and often complains of chest pain. Chest x-ray shows bilateral hilar lymphadenopathy. Which of the following will indicate that the patient suffers from sarcoidosis?

A. Hemoptysis

B. Eosinophilia

C. Episcleritis

D. Erythema nodosum

E. Cranial nerve XII dysfunction

15. A 43-year-old patient presented to ER with nasal bleed stuffy nose, also complaints of joint pains, he has a history of saddle-nose deformity. X-ray chest reveals nodular infiltrates. Which of the following is the induction therapy for patients with Wegner's granulomatosis?

A. Methotrexate plus steroids

B. Azathioprine plus steroids

C. Cyclophosphamide plus steroids

D. Trimethoprim plus steroids

E. Rituximab plus steroids

A 35-year-old female presents to ER with fatigue, sleep disturbance, weakness, and muscle spasms. Her blood pressure was 130/80 mmHg, pulse 74/minute, respiratory rate 17/minute. In fibromyalgia, all of the following tests are indicated except?

- A. CRP level
- B. Hemoglobin level
- C. WBC Count
- D. TSH level
- E. Coagulation function test.

17. A 28-year-old female presented to ER with migraine and lace-like purplish rash on the skin. She has a history of recurrent miscarriages. Laboratory tests that reveal lupus anticoagulants are in high titers. Which of the following will confirm that the patient is suffering from antiphospholipid syndrome?

- A. Evidence of stroke
- B. Evidence of DVT
- C. Elevated lupus anticoagulant is enough to establish the diagnosis
- D. Elevated antiphospholipid antibody after 12 weeks
- E. Fetal loss before the 10th week of pregnancy

18. Which of the following is true regarding reactive arthritis?

- A. More frequent in the upper extremities
- B. It is a clinical diagnosis
- C. All patients are HLA -B27 positive
- D. It involves the joints symmetrically
- E. First-line therapy is with sulfasalazine

19. A 38-year-old man has had 3 months of intermittent joint pain and swelling in both knees. The symptoms last weeks at a time and then improve spontaneously. His only past medical history is that of an oval-shaped, red pruritic skin lesion in his right axilla 4 months earlier, which appeared shortly after a camping trip. His immunoglobulin G (IgG) serology for *Borrelia burgdorferi* is positive. Which of the following is the most appropriate next step in management?

- A. High-dose glucocorticoids
- B. Low-dose glucocorticoids

C. High-dose NSAID therapy

D. Long-term penicillin

E. Long-term doxycycline

20. A 50-year-old man swims daily in a lake. Approximately 2 years ago he injured his left foot while in the lake after stepping on some rocks. Pain and swelling developed at the site of injury, which was just proximal to the 3rd digit on his left foot. Traumatic synovitis was diagnosed, and he was started on oral ibuprofen and local corticosteroid injections. During the next 2 years, the pain and swelling have persisted, and now have spread to the plantar surface of his foot and to his ankle. He has not had any fever or chills. Three synovectomies have been performed, and each showed non-caseating granulomas. Special stains revealed no mycobacteria or fungi. Cultures have not grown anything at 2 months since the last synovectomy. He comes to you today because it is worsening again. Which of the following organisms is most likely causing his infection?

A. *Mycobacterium marinum*

B. *Nocardia brasiliensis*

C. *Mycobacterium tuberculosis*

D. *Blastomyces dermatitidis*

E. *Sporothrix schenckii*

21. A 28-year-old woman presents to the emergency department because of severe pain and swelling in multiple joints. Two weeks ago, she was admitted with severe gastrointestinal dysentery, from which she has recovered. Ibuprofen has offered little relief. Physical examination reveals overt erythema, warmth, and swelling of the fingers, hands, wrists, and elbows. She has a swollen left knee. Laboratory testing reveals an ESR 88 mm/hr and CRP 182 mg/L (normal <5). All of the following microorganisms are associated with reactive arthritis except:

A. *Staphylococcus aureus*

B. *Streptococcus pyogenes*

C. *Chlamydia trachomatis*

D. *Yersinia pestis*

E. *Salmonella enteritidis*

22. Masha is a 31-year old female from Mafia who has been attending a peripheral health facility for her skin disease since 5 years ago. She has been cared for by topical medications. She has been taking on and off paracetamol for a joint disease she noticed since last year. She reports that the condition has been worsening, making her to seek referral to Muhimbili National Hospital. You happen to see the patient and have the opportunity to suggest a pharmacological agent.



Which of the following should be avoided in this patient?

- A. Methotrexate
 - B. Prednisolone
 - C. Sulfasalazine
 - D. Azathioprine
 - E. Leflunomide
23. When attending a 50-year-old patient at Rheumatology clinic, two residents agreed that the patient had dactylitis. In sorting out the most likely diagnosis, which of the following statements is true?
- A. Osteoarthritis is likely
 - B. Gout is the most likely diagnosis
 - C. Psoriatic arthritis is likely
 - D. Rheumatoid arthritis and psoriatic arthritis are both likely
 - E. It is the commonest finding in reactive arthritis

24. Chihwaga is a 29-year-old man with history of skin psoriasis since age 12. He presents with severe low back pain with stiffness that lasts for more than an hour in the morning. Pain improves with activity and stretching. MRI shows acute, bilateral sacroiliitis. He is currently on topical medications for skin psoriasis, but still has a psoriasis body surface area of 2%. He is currently on NSAIDs, which have only partially improved his back pain. He was referred to rheumatology for further evaluation. What is the next step in management?

- A. Start methotrexate
- B. Start leflunomide and sulfasalazine
- C. Continue NSAIDs and physical therapy
- D. Start Prednisone PO
- E. Start infliximab

25. Viral Hemorrhagic syndrome is based on vascular instability and decreased vascular integrity as a result of:

- A. Direct or indirect assault on the microvasculature
- B. Unexplained increased vascular permeability
- C. Local hemorrhage
- D. Decreased blood pressure and severe shock

26. A 36-year-old woman is reviewed. She presented 4 weeks ago with itchy dry skin on her arms and was diagnosed as having atopic eczema. She was prescribed hydrocortisone 1% cream with an emollient. Unfortunately, there has been no improvement in her symptoms. What is the next step in management, alongside continued regular use of an emollient?

- A. Betamethasone valerate 0.1%
- B. Clobetasone butyrate 0.05%
- C. Clobetasol propionate 0.05%
- D. Topical tetracycline
- E. Regular wet wraps

27. A 64-year-old woman presents with severe mucosal ulceration associated with the development of blistering lesions over her torso and arms. On examination the blisters are flaccid and easily ruptured when touched. What is the most likely diagnosis?

- A. Pemphigus vulgaris

- C. Bronchiectasis
- D. Necrotizing vasculitis
- E. Pleural effusion

32. A 10-year-old child has recurrent signs and symptoms of palpable purpura on the buttocks, arthralgias, colicky abdominal pain, diarrhea, and microscopic hematuria. Which of the following is the most likely diagnosis?
- A. Influenza
 - B. Immune complex vasculitis
 - C. Juvenile RA
 - D. systemic lupus erythematosus (SLE)
 - E. Wegener's granulomatosis
33. Which of the following conditions is least likely to exhibit the Koebner phenomenon?
- A. Vitiligo
 - B. Molluscum contagiosum
 - C. Lichen planus
 - D. Psoriasis
 - E. Lupus vulgaris
34. A 45 years old male presents with upper right sided abdominal pain for the past 6 months. On examination he is icteric, with gross ascites and a ballotable liver. Investigations indicate that Hepatitis B surface antigen is positive, he has markedly deranged liver function, alpha-fetoprotein is 1000ng/mL. Triphasic computerized tomography of the abdomen reveals multiple nodules that are hypervascular on arterial phase with rapid washout in Porto venous phase. What is the most likely mechanism for his disease?
- A. Chronic inflammation and regeneration
 - B. Overexpression of hepatitis B slice protein
 - C. Hepatitis B DNA integration into host genome
 - D. Interaction of hepatitis B X protein with host p53
 - E. Increased production of truncated pre S2/S protein
35. A 40 years old male is referred for care on account of a positive Hepatitis B surface. His wife is negative for Hepatitis B surface antigen. The wife requests if she can be

immunized. What additional
Hepatitis B immunization?

- A. Hepatitis B core IgM
- B. Hepatitis B core IgG
- C. Hepatitis B viral DNA
- D. Hepatitis B surface antigen
- E. Hepatitis B surface antibodies

36. A 45 years old male admitted at Mwaisela ward 6 presents on account of acute with hematemesis and hematochezia. He informs that he was diagnosed with Hepatitis B 5 years ago. Which of the following interventions would provide better outcomes?

- A. Octreotide
- B. Band ligation
- C. Adrenaline 1:10,000
- D. Adrenaline 1:100,000
- E. Octreotide and band ligation

37. A 40 years old male was noted to be positive for HBsAg and HBeAg during a blood donation campaign and is referred to you for further work up. His ALT and AST are within normal parameters. What is your next step?

- A. Liver biopsy
- B. Interferon therapy
- C. Observation
- D. Hepatitis-DNA estimation
- E. Check for serum albumin

38. A patient presents with fever and dry cough for the past 10 days full blood picture indicates that the white cell counts are $2 \times 10^3/\mu\text{L}$. The patient is likely to be infected by which of the following organisms?

- A. *Streptococci pyogenes*
- B. *Coccidiomycosis*
- C. *Candida species*
- D. *Sarcoptes scabiei*
- E. *Isospora belli*

38. A 59 years old patient known to be asthmatic presents with productive cough anorexia and fever for the past 3 days. You examine her and note that she has rhonchi, temperature of 39°C . What will you do to confirm the diagnosis?

- A. CT scan of the chest
- B. Chest radiograph
- C. Sputum for acid fast bacilli
- D. Aspergillus skin testing and measurement of Ig E levels
- E. Bronchoscopy

40. A patient presented with cough with purulent tenacious sputum, fever and difficulty in breathing for the past 10 days. His FEV1/FVC is reduced. A CT scan is performed see image below. The clinician at emergency medicine performs a chest radiograph and notes.



The best option for prevention will include:

- A. Daily Ciprofloxacin 500mg twice a day
- B. Daily ampicillin 1gm thrice a day
- C. Cloxacillin 1 gm thrice daily
- D. Isoniazid 100mg daily for 6 months
- E. Rifampentine for 6 months

41. A 48 years old patient admitted at the Intensive care unit for the past 7 days on account of a cerebrovascular accident and has been on a ventilator ever since. As you pass rounds you note that there is increased tracheal secretions his temperature is 39°C and he is noted to be hypoxemic. What is your next step?

- A. Cover the patient with ceftriaxone 1gm twice daily
- B. Cover the patient with meropenem 2gm once daily
- C. Cover the patient with meropenem 2gm twice daily
- D. Piperacillin 3gm 4 hourly

E. Daptomycin 250mg once daily

42. A 25 years old patient otherwise healthy is admitted in hospital on account of closed fracture to his left femur later developed pulmonary embolism and is admitted to the intensive care unit and is on a ventilator for the past 72 hours. You are called to review him and you note the following temperature 38.2°C , you note that there are non-purulent tracheal secretions, his $\text{PaO}_2/\text{FIO}_2$ is 250 mmHg. White cell counts were 6,000 uL. portable chest radiograph indicates a diffuse infiltrate and when you compare with the previous radiograph there is no progression. There was moderate growth on tracheal aspirate. What would be your line of thinking?

- A. The patient's findings are suggestive of hospital acquired pneumonia
- B. The patient is less likely to have hospital acquired pneumonia
- C. The patient might have malaria
- D. The patient has community acquired pneumonia
- E. The patient femur fracture has complicated

43. A young man with tuberculosis presents with massive hemoptysis that is uncontrollable. What would be your next plan of action?

- A. Angiographic treatment following evaluation of the aorta
- B. Angiographic treatment following evaluation of the pulmonary artery
- C. Angiographic treatment following evaluation of the bronchial artery
- D. Angiographic treatment following evaluation of the pulmonary vein
- E. Angiographic treatment following evaluation of the superior vena cava

44. A 19-year-old, asymptomatic female has missed her period for three consecutive months. A urinary pregnancy test turns out positive. Her urine is cloudy with culture growing 100,000 CFU of a gram-negative rod. Which of the following is appropriate treatment for her?

- a. Ciprofloxacin 500 mg twice daily for 5 days
- b. Trimethoprim-sulfamethoxazole 1 tab twice daily for 2 weeks
- c. Nitrofurantoin 100 mg daily for 3 days
- d. Ceftriaxone 1g IV daily for 2 weeks
- e. Encourage to drink lots of fluids

45. A 23-year-old female presents with headache, malaise, anorexia, pain in both sides of her jaws and discomfort in both lower abdominal quadrants. On examination she is febrile, tachycardic, both parotid glands are enlarged, bilateral lower abdominal tenderness. What is the most appropriate treatment for this condition?

- a. Symptomatic
- b. Immunization
- c. Broad spectrum antibiotics
- d. Steroids
- e. Sulphonamides

46. A female patient reports that she over saw a bottle of HIV antiretrovirals in her husband's closet and is worried that he might be HIV infected and is taking medications secretly. You decide to test her and find that she is not reactive to the rapid test. She reports that the last coitus was 6 months ago and she has been worried after seeing the antiretrovirals that seems to be reducing in number at an irregular pace. The coital acts with her spouse are unplanned. What would be your next best step

- A. Provide her with daily lamivudine, dolutegravir and tenofovir
- B. Provide her with daily emtricitabine, dolutegravir and tenofovir
- C. Provide her with daily emtricitabine, efavirenz and tenofovir
- D. On demand tenofovir and emtricitabine
- E. Daily tenofovir and emtricitabine

47. A 65 years old patient is known to be HIV infected come to you with lesions that resemble oral Kaposi's sarcoma in the mouth, extremities and tongue and has been on and off several anti-retroviral treatment regimens at times he buys medications and stops due to stigma. He cannot tell you what medications he has been on as he cannot remember the names. You decide to do HIV resistant testing and note that he has K65R mutation. What drug would be useful to consider for his treatment?

- A. Tenofovir
- B. Abacavir
- C. Zidovudine
- D. Didanosine
- E. Stavudine

8. A 34-year-old man with a history of polyarthralgia, back pain and diarrhea is found to have a 3 cm red lesion on his shin which is starting to ulcerate. What is the most likely diagnosis?



- A. Systemic Shigella infection
 - B. Syphilis
 - C. Metastatic colon cancer
 - D. Erythema nodosum
 - E. Pyoderma gangrenosum
49. YPD is a 30 years old male with a three days history of fever, productive cough, chest pain on taking in breaths and loss of appetite. On examination he is fully conscious oriented to person time and place with a respiratory rate of 32 breath per minute and a blood pressure of 85/60 mmHg. His White cell count are $15 \times 10^3/\mu\text{L}$, urine analysis no protein, Creatinine-80mmol/L, Urea 3 mmol/L. The intern makes a diagnosis of pneumonia and is unsure of the severity and management plan. You will advise the intern that
- A. The pneumonia is mild admit in the patient to the medical ward
 - B. The pneumonia is moderate admit the patient to the medical ward
 - C. The pneumonia is moderate treat as an outpatient
 - D. The pneumonia is severe admit to the intensive care unit
 - E. The pneumonia is severe admit to the high dependent unit

50. A 45 years old male is HIV infected presents with a dry cough, difficulty in breathing but dies before any investigation. You request a post-mortem as you were concerned that he might have died of pneumocystis jirovecii pneumonia. Which statement is the most correct?

- A. Prominent interstitial pneumonitis
- B. Eosinophilic alveolar exudates
- C. Prominent mononuclear cell in alveolar exudates
- D. Neutrophilic infiltration of the alveolar interstitium
- E. Hypertrophy of type I pneumocytes

51. A 45 years old male patient presents to the ward with history of lethargy, high grade fever and cough. He reports that the sputum is chocolate in color. Which organism will you be concerned of?

- A. Legionella
- B. Neisseria meningitis
- C. Streptococcus pneumonia
- D. Klebsiella
- E. Mycoplasma

52. An 85-year-old lady presents to dermatology clinic complaining of itchy white plaques affecting her vulva. There is no history of vaginal discharge or bleeding. A similar plaque is also seen on her inner thigh. What is the likely diagnosis?

- A. Candida
- B. Lichen planus
- C. Lichen sclerosus
- D. Herpes simplex
- E. Seborrheic dermatitis

53. A 25 years old male presents with painful urination and pain in the urethra. He reports a clear penile discharge. Urethral pain is worse in the mornings. He also has pain on defecation and a sense of incomplete bowel emptying. You plan to investigate him. Which statement is the most correct?

- A. Collection of mid-stream mid -day urine with at least 3 WBC/high power field is diagnostic

- B. Collection of first voided morning urine with at least 5 WBC /high power field is diagnostic
 - C. Collection of terminal spot urine with at least 5 WBC /high power field is diagnostic
 - D. Collection of first voided mid -day urine with at least 3 RBC/high power field is diagnostic
 - E. Collection of mid- stream urine at mid -day with at least 5 RBC/high power field is diagnostic
54. A 26 years old male presents to the out- patient clinic with a painless ulcer on the penile shaft that has been there for the past 8 weeks. You examine him and note that the ulcer has irregular margins with an indurated base and serous discharge. His inguinal lymph nodes are enlarged and non-tender. Which is the most correct statement
- A. Venereal Disease Research Laboratory Test is highly specific for the disease
 - B. Rapid plasma regain is highly specific for the disease
 - C. Culture for Chlamydia is highly specific
 - D. Culture for Treponema Pallidum is highly specific
 - E. Dark field microscopy is highly specific
55. A 23 years old male presented with a painless penile ulcer that was treated benzathine penicillin injection after six hours he began experiencing fever, headache, sweating rigors. Which statement is most correct?
- A. He has developed a drug hypersensitivity reaction
 - B. He is likely undergoing a cytokine storm
 - C. The symptoms are likely to continue for a week
 - D. This is an unexpected presentation
 - E. He should be investigated for malaria
56. A male patient presents with extensive visceral Kaposi's Sarcoma despite being on highly active antiretroviral therapy. What would be your next best option
- A. Radiotherapy
 - B. Lymph node excision
 - C. Intralesional vinblastine
 - D. Liposomal doxorubicin
 - E. Interferon alpha

57. A 43-year-old man comes for review. A few months ago, he developed redness around his nose and cheeks. This is worse after drinking alcohol. He is concerned as one of his work colleagues asked him if he had a drink problem despite him drinking 14 units per week. On examination he has erythema as described above with some pustules on the nose and telangiectasia on the cheeks. What is the most likely diagnosis?

- A. Mitral stenosis
- B. Seborrheic dermatitis
- C. Alcohol-related skin changes
- D. Acne rosacea
- E. Systemic lupus erythematosus

58. A 42-year-old woman is evaluated for a 2-month history of fatigue, tingling in the fingers of both hands, and pain radiating into the hands and forearms. She also has difficulty opening bottles. The tingling symptoms occasionally are alleviated when she shakes her hands in the morning, but she typically has 2 to 3 hours of morning stiffness in her wrists and fingers. She takes no medications. On physical examination, vital signs are normal. BMI is 21. Soft-tissue swelling and tenderness is palpable at the wrists and metacarpophalangeal joints bilaterally. She has sensory loss over the palmar surface of the first three digits and weakness of abduction and opposition of the thumbs bilaterally. Results from electrodiagnostic studies are consistent with bilateral median neuropathies of the wrist. Which of the following is the most likely cause of this patient's neuropathy?

- A. Osteoarthritis
- B. Overuse injury
- C. Polymyalgia rheumatica
- D. Rheumatoid arthritis

59. Which of the following is a hallmark symptom of bronchial asthma?

- A. Hemoptysis
- B. Chest pain
- C. Wheezing
- D. Epistaxis
- E. Fatigue

60. A patient with bronchial asthma presents with chronic symptoms and frequent exacerbations despite using a short-acting beta-agonist (SABA) as needed. What is the most appropriate step-up therapy?

- A. Adding an inhaled corticosteroid (ICS)
- B. Discontinuing the use of SABA
- C. Starting an oral leukotriene receptor antagonist (LTRA)
- D. Initiating oxygen therapy
- E. Prescribing an anticholinergic inhaler

- a. are viral in etiology
- b. may be premalignant lesions
- c. are found mainly in patients with lymphoma
- d. are contagious in children only
- e. may be treated with griseofulvin

30. A 22-year-old woman develops an acute contact dermatitis to a household-cleaning agent. Which of the following treatments is most appropriate during the bullous, oozing stage?

- a. wet dressings
- b. systemic corticosteroids
- c. topical anesthetics
- d. systemic antibiotics
- e. antihistamines

31. A 67-year-old man presents to the emergency department with symptoms of renal colic. Plain x-rays of the abdomen reveal no obvious renal stone. A CT scan of the abdomen is ordered to confirm the clinical diagnosis. Which of the following coexisting medical conditions increases the risk of contrast-induced nephropathy?

- a. Hyperparathyroidism
- b. Pyelonephritis
- c. Nephrolithiasis
- d. Renal cell carcinoma
- e. Multiple myeloma

32. A 62-year-old man with coronary artery disease (CAD) presents to the emergency room with symptoms of feeling faint. The symptoms started earlier in the day and are more prominent when he is standing up. He reports no chest pain or dyspnea and is otherwise well. Medical history includes type 2 diabetes, dyslipidemia, and hypertension. His medications are atenolol, metformin, atorvastatin, and ramipril. On physical examination, blood pressure is 125/70 mm Hg, heart rate is 56/min and irregular. Heart sounds S1 and S2 are normal, the lungs are clear, and he is alert and oriented. The electrocardiogram (ECG) shows Wenckebach type atrioventricular (AV) block. Which of the following are you most likely to see on the ECG?

- a. progressive PR shortening
- b. progressive lengthening of the PR interval
- c. tachycardia
- d. dropped beat after progressive lengthening of the PR interval
- e. fixed 2:1 block

33. A pulmonologist is testing a patient's lung volumes and capacities using simple spirometry. Which of the following lung volumes or capacities cannot be measured directly using this technique?

- a. Expiratory reserve volume
- b. Functional residual capacity
- c. Inspiratory reserve volume
- d. Tidal volume

- c. TDF/FTC/EFV
- d. TDF/FTC/NVP
- e. Peg-IFN

5. A 12 years Juma Walter, presents to you with fever, hemoptysis and cough which produces 180 mls of fowl smelling greenish sputum production per day for the past 2 months. His mother insist that Juma get recurrent cough since age of 3 years and each time it is associated with right sided chest pain. The following findings were noted on examination: Temperature = 38.9°C , wasting, finger clubbing, Pulse 120/min, BP=110/80mmHg. RR =40/min, central trachea, crepitations are noted right mammary and infra-mammary and right infra scapular areas.
- a. Juma's sputum culture must include anaerobic environment.
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6. The following are true regarding pleural effusion
- a. When Tuberculosis is suspected pleural fluid adenosine deaminase should be determined by Ziehl-Neelsen staining
 - b. Finding of high pleural fluid amylase and high pH suggest mesothelioma
 - c. Effusions of about 200ml can't be picked by a plain postero-anterior view of Chest X ray.
 - d. Ovarian malignancy is associated with right-sided pleural effusion.
 - e. Pleural effusion but not lung collapse will cause mediastinal shift.
7. A 27 years old female, presented with dysuria, increased micturition frequency and urgency. Physical examination reveals suprapubic tenderness and a yellow discharge was evident at her vaginal introitus. The following statements are TRUE regarding this patient's illness.
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 - e. Ampicillin or co-trimoxazole are the drugs of choice
8. A 65-year old patient presents to you with a painful right lower leg. On questioning, he tells you the pain has been present for a few months but he has not sought any medical advice for it. His past medical history is significant only for type 2 diabetes, for which he is on insulin. On examination he appears systemically well. Examination of the right lower limb shows a curvature of the tibia with overlying erythema, and

- ...activity indicates active disease state.
...ing of jaundice may follow treatment of HIV with HAART.
4. If you decide to start treating SG which of the following antivirals combinations would you consider in her initial combination?
 - a. ABC/3TC/EFV
 - b. AZT/3TC/LPV/r
 - c. TDF/FTC/EFV
 - d. TDF/FTC/NVP
 - e. Peg-IFN

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SEMESTER 2 SUPPLEMENTARY EXAMINATION (IM 601)

Date: 16/03/2020 Time: 2pm – 5 pm (3 hours)

INSTRUCTIONS

1. The paper consists of two sections A and B
2. ATTEMPT ALL questions in both section A and B.
3. Section A, consist of Multiple choice questions. Choose the **MOST CORRECT** answer and write the answer on the answer sheet attached to the question paper.
4. Section B, **MUST** be answered on the University answer booklets. Start each question on a **FRESH SHEET**. Make sure that same sheet does not contain two different questions in this section.
5. AVOID USE OF UNCONVENTIONAL ABBREVIATIONS

SECTION A: Multiple Questions: Choose the **MOST** correct answer (40 MARKS)

1. A 70- old lady is brought into the emergency department by her worried neighbours, who report her not being her usual self. On examination she looks unwell, confused and complains of a headache. She is febrile, has photophobia, resist flexion of the neck. What is the most likely organism involved?
 - a. Gram negative intracellular diplococcic
 - b. Acid fast bacilli
 - c. Alpha hemolytic streptococci
 - d. Coagulase positive staphylococci
 - e. Gram negative bacilli
2. The following infections can be prevented using vaccines
 - a. H1N1
 - b. HDV
 - c. HCV
 - d. V. cholerae
 - e. HBV
3. Mrs. SG is a 32 year HIV-infected patient diagnosed six months ago but not on antiretroviral therapy yet. She is seen for follow up in your clinic and noted to have a positive HbsAg both at this visit and one 6 months ago. Her CD4+ count dropped from 425 cells/mm³ at diagnosis to 275 cells/mm³ current visit. HIV RNA 100,000 IU/ml, HBV DNA is 3,000,000 IU/mL, ALT 23 IU
 - a. The overall risk of SG to develop chronic liver disease from chronic HBV if untreated is about 84%
 - b. Currently, liver damage or fibrosis for SG is unlikely.
 - c. Entecavir one of the drugs you would consider in her treatment.

23/10/18

12. A 40 year old man presented 14 days after return from a 6-week field trip to Papua New Guinea. He had a six day history of high fevers and rigors. On the day of presentation, he had become vague and confused. He had taken antimalarials as prophylaxis, but ceased when he found that local people did not take them. His temperature was 40°C, pulse rate 140bpm, respiratory rate 28 per minute, and blood pressure 100/60 mmHg. He had dry mucous membranes, mild jaundice, and blood splenomegaly and generalized crackles in both lungs. Full blood examination revealed: 6.5g/dl, WCC $2.5 \times 10^9/L$, Platelets $10 \times 10^9/L$, bilirubin $60 \mu\text{mol/L}$ (3-20), Lactate dehydrogenase 489 U/L (100-225), creatinine $250 \mu\text{mol/L}$. What is the likely diagnosis?

- Leishmaniasis
- Tick bile fever
- Endocarditis
- Falciparum malaria
- Viral hemorrhagic fever

13. A 62 year old lady presents with fever and persistent difficulty in speaking. Her signs show a temperature of 39°C. The patient was alert and oriented with respect to time but was unable to name objects properly. Dysarthria and occasional word substitution were noted. The patient followed two but not three-step commands. A provisional diagnosis of an aphasic temporal lobe lesion was made. A CT scan showed a low attenuation lesion involving the medial and posterior aspect of the left temporal lobe and inferior basal ganglia. What is the diagnosis?

- Polymorpho leukoencephalopathy
- Cerebral toxoplasmosis
- Herpes simplex encephalitis
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14. A 41 year old Caucasian female presented with malaise, dysphagia and sclerodactyly and Raynaud's phenomenon for the last 3 months. On physical examination she was afebrile and had a supine blood pressure of 110/80mmHg. Sclerodactyly and telangiectasia were observed in both hands. Blood tests revealed: Hb 11.5g/dl, MCV 85fl, ESR 80mm/hr, ANA strongly positive, anti-topoisomerase-I antibody (formerly anti SCL-70 antibody) positive, normal C3 and C4, anti-DNA, anti-centromere, anti-RNP, anti-Ro and La antibodies - negative, chest x ray showed bilateral basilar interstitial infiltrates. What is the diagnosis?

- Hereditary hemorrhagic telangiectasia
- Sjogren's syndrome
- Wegener's granulomatosis
- Oesophageal carcinoma
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15. A 22 year old woman presented to the emergency department because of progressive severe unilateral knee pain of 24 hour duration. There was no preceding injury or unusual physical activity was identified. She did not have any significant past medical history, but has been on a trip around Europe over the last month. Plain radiographs of

palpation reveals it to be much warmer than the left leg. Which one of the following is the most likely diagnosis?

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- Paget's disease
- Sarcoma
- Skeletal metastasis
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9. A 58 year old man has presented with chest pain and pulmonary oedema. He is managed as acute coronary syndrome. He did not tolerate a GTN infusion as his blood pressure was 85/60 but improved with frusemide and his blood pressure stabilized. An ACE-inhibitor was held off due to renal impairment. He is coincidentally found to have a chest infection, his blood cultures grew streptococci and he was treated with amoxicillin/clavulanic acid. His blood tests on admission show a creatinine of 145 $\mu\text{mol/l}$ rising up to 190 $\mu\text{mol/l}$ and then 250 $\mu\text{mol/l}$ subsequent two days and acute kidney injury is established. What is the likely cause of acute kidney injury?

- Urinary tract infection
- Acute tubular necrosis
- Interstitial nephritis due to the antibiotic
- Post streptococcal glomerulonephritis
- Renal artery stenosis

10. A 55 year homeless man was found collapsed at home and brought to the emergency department. He was unable to give a history. On examination, he smelled of alcohol. Examination reveals: temperature of 34°C, a pulse of 45bpm, pressure of 110/80mmHg. Dipstick urine analysis shows blood +++, protein +, glucose -ve. Some of his investigation results are listed below: Sodium 135mmol/l, potassium 4.5mmol/l, urea 5mmol/l, creatinine 300 $\mu\text{mol/l}$, AST 320 U/L, LDH 800 U/L. What is the likely cause of the raised serum creatinine concentration?

- Chronic renal failure
- Dehydration
- Hypothermia
- Paracetamol poisoning
- Rhabdomyolysis

11. A 2 year old girl presented with a 12-hour history of fever and poor feeding. The family owned two cats. Her temperature was 39.5°C, and she was irritable, with no localizing signs or skin lesions. A full septic screen was performed. Cerebrospinal fluid showed a neutrophilic pleocytosis and gram-negative coccobacilli. She was treated with intravenous cefotaxime and gentamicin. Within 24 hours both CSF and blood cultures showed growth of gram-negative bacilli. What is the likely infective organism?

- Legionella pneumophila
- Haemophilus influenza
- Pasteurella multocida
- Pseudomonas aeruginosa
- Neisseria meningitidis

- c. Lichen planus
- d. Secondary syphilis
- e. Pityriasis rosea

20. A 29 year old female presents to her GP. She had noticed dandruff and rash on her face. On examination there is evidence of areas of erythema with fine scaling over her nasal bridge, around her nasolabial fold and eyebrows and ears and there is evidence of fine scaling on the scalp. What is the most likely diagnosis?

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- b. Psoriasis
- c. Rosacea
- d. Eczema
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21. A 28 year old soldier presents with itchy rash. He is itchy in his finger webs, wrists and groin and there is evidence of excoriation in these areas. What is the most likely diagnosis?

- a. Contact dermatitis
- b. *Sarcoptes scabiei* infection
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- d. Lichen Planus
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22. A 45 year old office worker presented having felt unwell since returning from a 4-week holiday in Egypt. On examination, he was jaundiced. Investigations: HBsAg not detected, anti-hepatitis A virus IgM detected, anti-hepatitis A virus IgG not detected, anti-hepatitis C virus antibody not detected. For how long should he be excluded from working in his office?

- a. No exclusion necessary
- b. Until 7 days from onset of jaundice
- c. Until 14 days from onset of symptoms
- d. Until asymptomatic
- e. Until faecal samples are negative for hepatitis A virus

23. A 22 year old Indian student was found to have pulmonary tuberculosis. He was treated with rifampicin, isoniazid, pyrazinamide and ethambutol. What is the mechanism of action of rifampicin in the treatment of mycobacterial infection?

Inhibits synthesis of mycolic acid in mycobacterial cell wall

- a. Increases mycobacterial protein synthesis
- b. Inhibits mycobacterial RNA polymerase
- c. Inhibits mycobacterial fatty acids
- d. Inhibits synthesis of mycolic acid in mycobacterial cell wall
- e. Inhibits synthesis of mycolic acid in mycobacterial cell wall

24. A 35 year old presents with red scaly patches on her face, scalp and hands. On examination there is evidence of red plaques on her face, hands and scalp with some

the knee were unremarkable. On examination, the left knee, right knee, right wrist and elbows were erythematous and swollen. She did not have a skin rash. A joint aspiration was done and it showed no organisms but had increased white cells. She did not show any cultures. Her ESR is 60mm/hr. A medical social worker was called and the following information was obtained:

- a. Lyme disease
- b. Reactive arthritis
- c. Septic arthritis
- d. Psoriatic arthritis
- e. Gonococcal arthritis

16. A 33 year old female with systemic lupus erythematosus has arthritis involving her upper limbs. She also has a butterfly facial rash and a rash on the trunk. Urine dipstick shows no proteinuria or hematuria. Her renal function is normal. Which one of the following medications is most appropriate?

- a. Methotrexate
- b. Prednisolone
- c. Azathioprine
- d. Hydroxychloroquine
- e. Cyclosporine

17. A 55 year old woman, originally from Thailand, presented with significant weight loss and was found to have HIV infection. Investigations: HIV viral load 156250copies/ml (lower detection limit 40/ml), CD4 count $285 \times 10^6/L$ (430-1690), HBsAg positive, HBeAg negative, anti-hepatitis B core antibody positive, HBV DNA 950 IU/ml (lower detection limit 250). What is the most appropriate antiretroviral therapy?

- a. Abacavir/lamivudine and efavirenz
- b. Abacavir/lamivudine and nevirapine
- c. Tenofovir/emtricitabine and nevirapine
- d. Tenofovir/emtricitabine/and nevirapine
- e. Zidovudine/lamivudin and nevirapine

18. A 27 year old woman presented with a short history of general malaise, increased frequency of passing urine and dysuria. She was attempting to conceive and was reluctant to take any antibiotic that could be harmful in pregnancy. What antibiotic is most appropriate?

- a. Cefalexin
- b. Ciprofloxacin
- c. Cotrimoxazole
- d. Doxycycline
- e. Trimethoprim

19. A 22 year old presents with a rash. On examination there are small salmon pink papules with a fine scale which are drop like on her trunk and proximal arm and legs. She had suffered from tonsillitis two weeks previously. What is the most likely diagnosis?

- a. Guttate Psoriasis

A 25-year-old female resident comes to emergency department after a neck injury which happened about 30 minutes prior to the consultations while attending an emergency laparotomy following a perforated duodenal ulcer. She is crying and worried that she might lose her "wedding planned to happen in 6 months". With reason discuss how you will manage this case. (15 marks)

2. Discuss the management of a 17 years old school boy presenting with 1 mmHg measured on three different occasions (separate days) (15 marks)

3. Write short notes on the following

- Differential diagnosis and diagnostic work out of 20 years male right knee swelling for 5 days. His other joints are all normal
- Management of hyperkalemia (5 marks)

4. Match item in column A with that in Column B (10 marks)

A

B

- | | | |
|---|-------|---------|
| 1 | _____ | CREST |
| 2 | _____ | Grotons |
| 3 | _____ | Koebner |
| 4 | _____ | ddNA |

- | |
|--------------------|
| SLE |
| Systemic sclerosis |
| Dermatomyositis |
| Lichen planus |

1. A 25-year-old female resident comes to emergency department after a needle stick injury which happened about 30 minutes prior to the consultations while she was attending an emergency laparotomy following a perforated duodenal ulcer. She is crying and worried that she might lose her "wedding planned to happen in 3 weeks". With reason discuss how you will manage this case. (15 marks)

2. Discuss the management of a 17 years old school boy presenting with BP 180/120 mmHg measured on three different occasions (separate days) (15 marks)

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4. Match item in column A with that in Column B (10 marks)

A

B

... which happened about 20 minutes prior to the consultation while I
 attending an emergency operation following a perforated appendix
 crying and worried that she might have her "wedding planned" to happen
 With recent divorce here you will manage the case (15 marks)

Describe the management of a 17 year old child who presenting with 20
 mostly occurred in these different manner (approx 20 marks)

Write short notes on the following

- Differential diagnosis and management with one of 20 years with
 right knee swelling for 2 days. The other joints are all normal (10 marks)
- Management of hypernatremia (10 marks)

Match item in column A with item in Column B (10 marks)

A

B

1. Cystitis
2. Gonorrhoea
3. Syphilis
4. HIV

- a. B
- b. Gonorrhoea
- c. Syphilis
- d. HIV

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A

1 _____ CREST

B

SLE 4

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30. A 22-year-old woman develops an acute contact dermatitis to a household cleaning agent. Which of the following treatments is most appropriate?
- wet dressings
 - systemic corticosteroids
 - topical anesthetics
 - systemic antibiotics
 - antihistamines
31. A 67-year-old man presents to the emergency department with symptoms of renal colic. Plain x-rays of the abdomen reveal no obvious renal stone. A CT scan of the abdomen is ordered to confirm the clinical diagnosis. Which of the following coexisting medical conditions increases the risk of contrast-induced nephropathy?
- Hyperparathyroidism
 - Pyelonephritis
 - Nephrolithiasis
 - Renal cell carcinoma
 - Multiple myeloma
32. A 62-year-old man with coronary artery disease (CAD) presents to the emergency room with symptoms of feeling faint. The symptoms started earlier in the day and are more prominent when he is standing up. He reports no chest pain or dyspnea and is otherwise well. Medical history includes type 2 diabetes, dyslipidemia, and hypertension. His medications are atenolol, metformin, atorvastatin, and ramipril. On physical examination, blood pressure is 125/70 mm Hg, heart rate is 56/min, and irregular. Heart sounds S1 and S2 are normal, the lungs are clear, and he is alert and oriented. The electrocardiogram (ECG) shows Wenckebach type II atrioventricular (AV) block. Which of the following are you most likely to see on the ECG?
- progressive PR shortening
 - progressive lengthening of the PR interval
 - tachycardia
 - dropped beat after progressive lengthening of the PR interval
 - fixed 2:1 block

33. A pulmonologist is testing a patient's lung volumes and capacities using simple spirometry. Which of the following lung volumes or capacities cannot be measured directly using this technique?

- Expiratory reserve volume
- Functional residual capacity
- Inspiratory reserve volume
- Tidal volume

bald patches. Some of the patches are scarring and discoloured. What is the diagnosis that fits best with this clinical picture?

- a. Chronic discoid lupus erythematosus
- b. SLE
- c. Psoriasis
- d. Seborrhoeic dermatitis
- e. Drug induced lupus

25. A 72 year old female presents to her GP after noticing blisters on her arms and legs. She has recently been commenced on furosemide. On examination there is tense blisters evident predominantly on the flexural surfaces of her arms and legs. There is no evidence of blisters in her mucous membranes. There is some healing areas where the blisters have burst but no evidence of scarring. A skin biopsy is organized and reveals subepidermal blister and inflammatory infiltrate with numerous eosinophils. What is the most likely diagnosis?

- a. Bullous Pemphigoid
- b. Pemphigus vulgaris
- c. Pyoderma gangrenosum
- d. Dermatitis herpetiformis
- e. Eczema herpeticum

26. Which of the following symptom is typical of a patient with hypercalcaemia?

- a. Paraesthesia
- b. Dyspepsia
- c. ✓ Seizures
- d. Polydipsia
- e. Dysuria

27. Which of the following bacteria is the most common cause of UTI encountered in general practice?

- a. Klebsiella
- b. Enterobacter
- c. Enterococcus
- d. Proteus
- e. ✓ E. Coli

28. A 25-year-old woman attended the melanoma clinic with her father, aged 53, whose history included two melanomas. They both had the atypical mole syndrome. What factors carries the strongest risk of her developing melanoma?

- a. Atypical mole syndrome
- b. Blistering sunburn in childhood
- c. Corticosteroid use
- d. Sunbed use
- e. Type 1 skin

29. A 27-year-old man develops warts on his hand. Which of the following is a correct statement concerning these skin lesions?